

Python Notes

1. Introduction to Python

- Python is a high-level, interpreted programming language known for its simplicity and versatility.
- Features: Easy syntax, vast libraries, cross-platform, open-source.
- Used in: Web development, data science, automation, AI, scripting.

2. Python Basics

- Syntax: Indentation-based structure.
- Comments: Use '#' for single-line, triple quotes for multi-line.
- Variables: Dynamically typed. Example: `x = 5`
- Data Types: int, float, str, bool, list, tuple, dict, set.
- Type Casting: `int()`, `float()`, `str()`, etc.
- Input/Output: `input()`, `print()`
- Operators: `+`, `-`, `*`, `/`, `%`, `**`, `//`

3. Control Flow

- if, elif, else for decision making.
- Loops: for, while.
- Loop control: break, continue, pass.

4. Functions

- `def function_name(params):`
- return statement returns values.
- Lambda: anonymous functions using lambda keyword.

5. Data Structures

- List: Ordered, mutable. Example: `my_list = [1, 2, 3]`
- Tuple: Ordered, immutable. Example: `my_tuple = (1, 2)`
- Dict: Key-value pairs. Example: `my_dict = {'a': 1}`
- Set: Unordered, no duplicates. Example: `my_set = {1, 2, 3}`

6. Object-Oriented Programming

- Class defines blueprint, object is instance.
- `__init__()` is the constructor.

- Inheritance: class Child(Parent)
- Encapsulation: hiding data using `_` or `__`
- Polymorphism: same method name with different behavior.

7. File Handling

- Open a file: `open('file.txt', 'r')`
- Read: `read()`, `readline()`, `readlines()`
- Write: `write()`, `writelines()`
- Use 'with' to auto-close files.

8. Error Handling

- try-except to catch exceptions.
- finally block always executes.
- raise keyword to throw exceptions.

9. Useful Modules

- math: mathematical functions.
- random: random number generation.
- datetime: work with dates and times.
- os: interacting with the operating system.
- sys: system-specific parameters and functions.

10. Practice Tips & Resources

- Practice on HackerRank, LeetCode.
- Build small projects like to-do apps, calculators.
- Follow official Python docs and tutorials.